

Swaps

Khagendra Gupta^{AC}

(44-20) 7134-0486

Khagendra.x.gupta@jpmorgan.com

J.P. Morgan Securities plc

Marcus Mayfield

(44-20) 3493-8050

Marcus.mayfield@jpmorgan.com

J.P. Morgan Securities plc

^{AC} Indicates certifying analyst. See last page for analyst certification and important disclosures.

All exhibits shown in the presentation are sourced from J.P.Morgan unless otherwise stated

Understanding Swaps: Class Objectives

- Learn how to intuitively understand and price a swap from first principles
 - Building a swap curve
 - €STR, 3s, 6s curve
- How to use swap curve as a common way to compare FI assets
 - Swap spreads
- Understanding commercial and structural reasons for using swaps
 - Myriad users of swaps
 - Pension funds, Insurance companies, Banks, etc

What are swaps?

- A swap is a **contractual agreement to exchange one set of cash flows for another, where the cash flows depend on some market variable**
- Swaps are therefore derivatives
 - A common Interest Rate Swap example - exchanging a set of fixed payments for payments linked to some floating (or short term) interest rate (such as Euribor/SOFR)
 - E.g., receiving 3.20% for 30Y versus paying floating Euribor for 30Y. This is equivalent to:
 - “selling” a floating rate savings account, and...
 - ...“buying” a 3.20% coupon 30Y bond, but...
 - ...without the exchange of assets & associated costs

Terminology: **Paying fixed vs. receiving floating**

- Pay fixed
- Short duration
- Payer
- **Borrowing** (Careful! No cash is exchanged)
- **Short swaps** (Ambiguous; don't use this!)

Terminology: **Receiving fixed vs. paying floating**

- Receive fixed
- Long duration
- Receiver
- **Lending**
- **Long swaps**

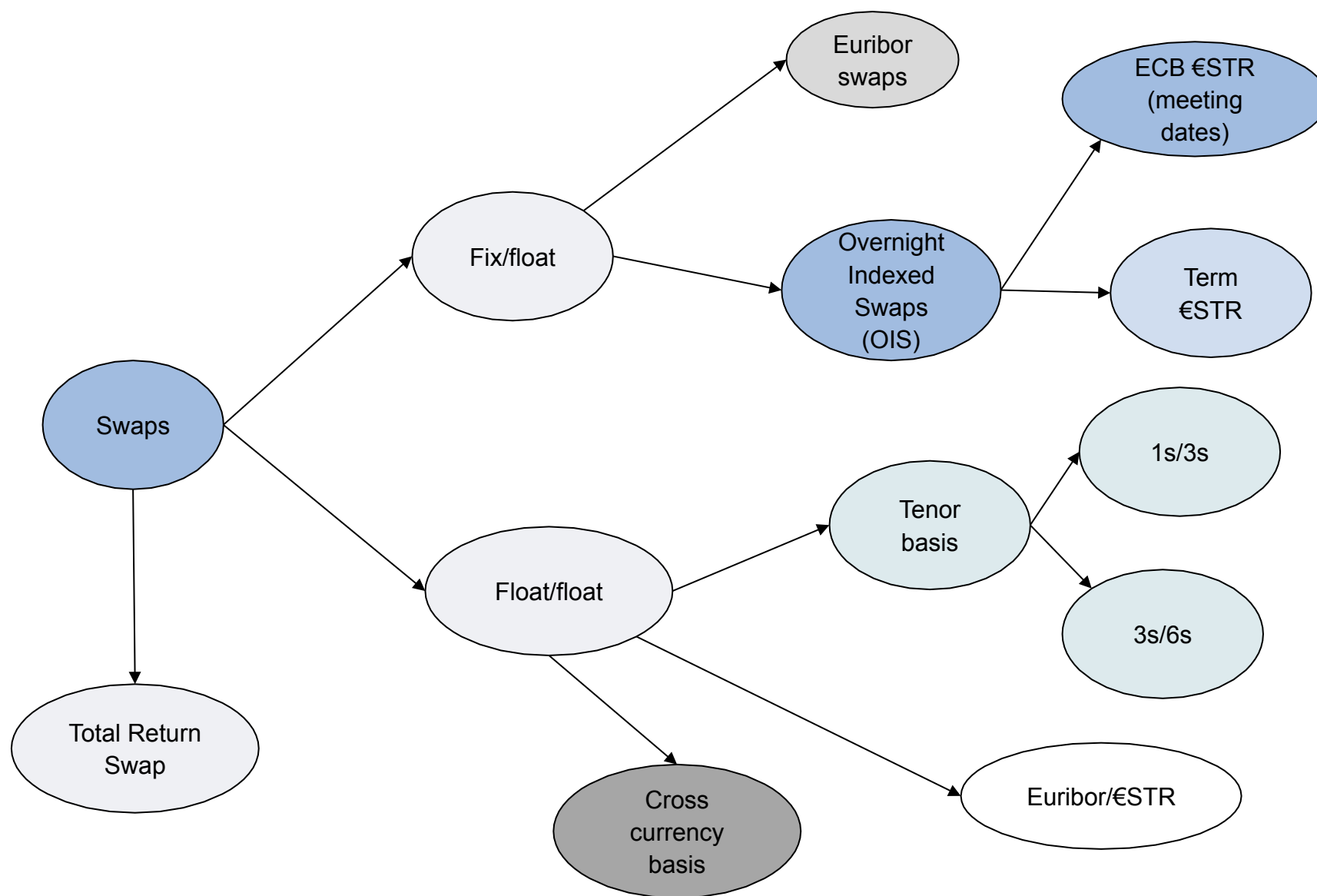
Why are swaps so popular?

- Self-funded, “no-money-down”
- Better liquidity relative to other asset classes
- Well-defined curve exists to long maturities
- Efficient way to keep Fixed/Floating debt markets aligned
- The swaps curve has become a benchmark curve, against which all other fixed income asset classes are valued
- Asset swaps are efficient ways to structure cross market relative value trades – comparing other assets against a common benchmark

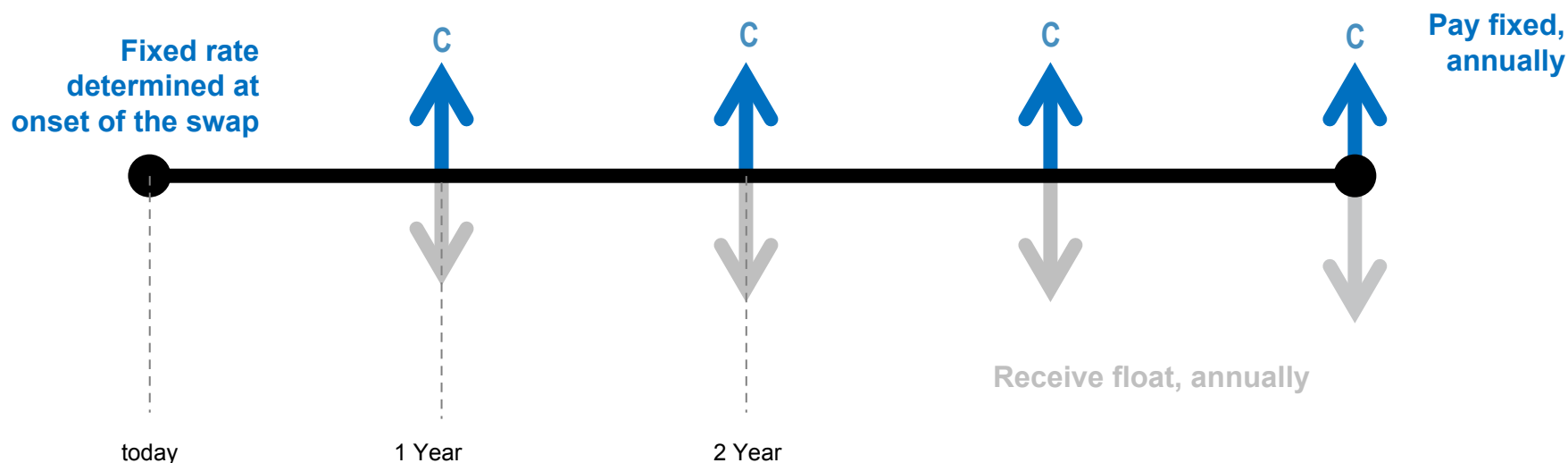
Uses of Swaps

- ***Manage duration and yield curve risk***
 - The size and the liquidity of the swaps market make it attractive for tactical duration and yield curve management
- ***Trade asset swap spreads***
 - Asset swap spreads on Treasuries and spread products alike offer attractive relative value/cross-market trading opportunities
- ***Create synthetic assets***
 - Buy short-duration assets and receive in an interest rate swap to create a synthetic fixed-rate bond

Type of EUR swaps



Basic structure of a swap



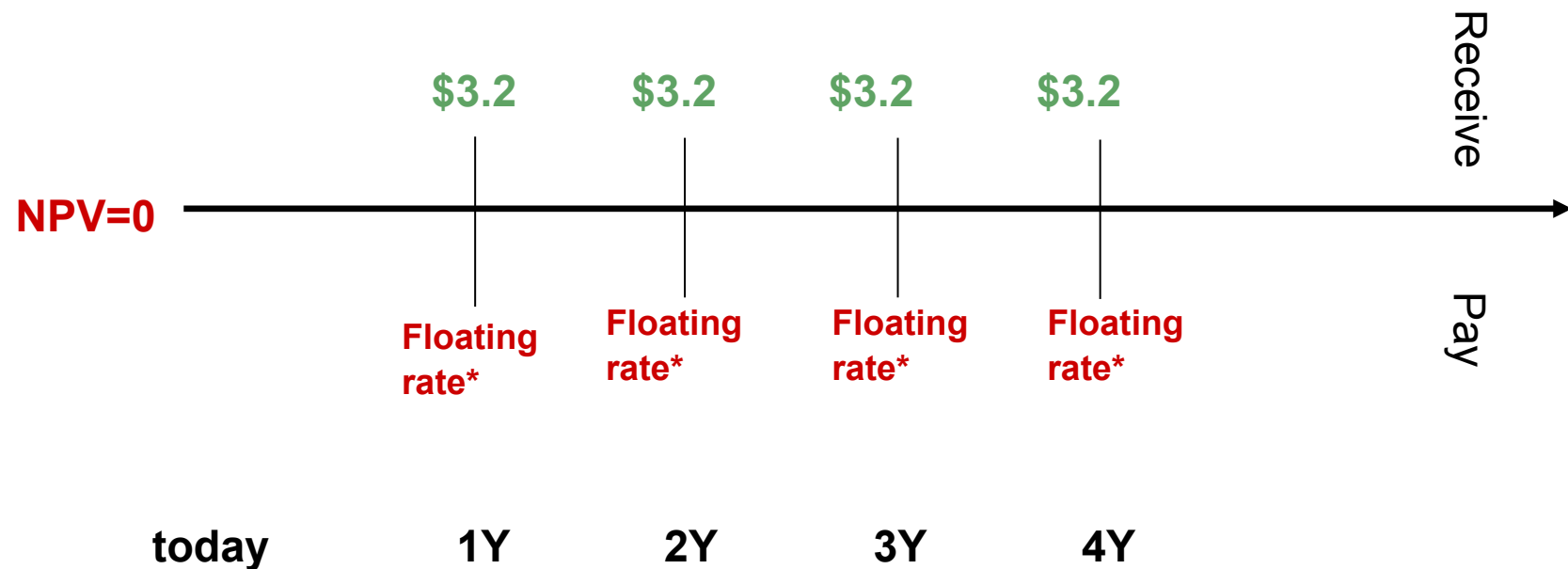
Standard, on market 2Y Euribor pay-fixed swap

- Fixed rate coupon paid annually (negative NPV)
 - For €STR swaps, 1-year €STR compounded in arrears, received annually (positive NPV);
 - Fixed rate is determined at the onset whereas floating rate is determined 1Y later (in arrears)
 - For Euribor swaps, 6M Euribor received semi-annually (positive NPV)
 - Both fixed & floating rates are determined at the onset; floating rate changes every 6M
- At inception, the NPV of the two legs are equal in magnitude (by design!)

currency	rate	settle	fixed leg		float leg	
USD	SOFR	T+2	Act/360	1Y	Act/360	1Y
EUR	ESTR (Euribor)	T+2	Act/360	1Y	Act/360	1Y (6M)
JPY	TONAR	T+2	Act/365	1Y	Act/360	1Y
GBP	SONIA	T+0	Act/365	1Y	Act/365	1Y
CHF	SARON	T+2	Act/360	1Y	Act/360	1Y

Swap Cash Flows

- Typical swap pays fixed leg and floating leg annually
- Fixed payments structured so that NPV=0....



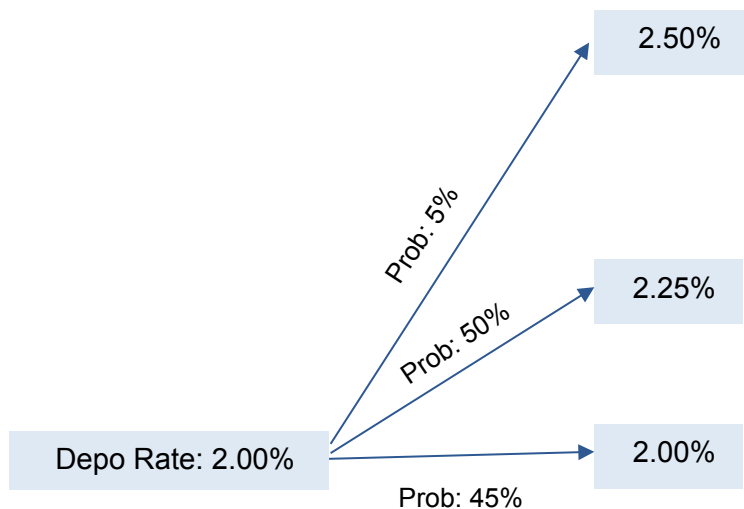
... but how did we find the fixed coupon (i.e., 3.2%) ?

*** Floating rate: SOFR for USD, typically Euribor for EUR**

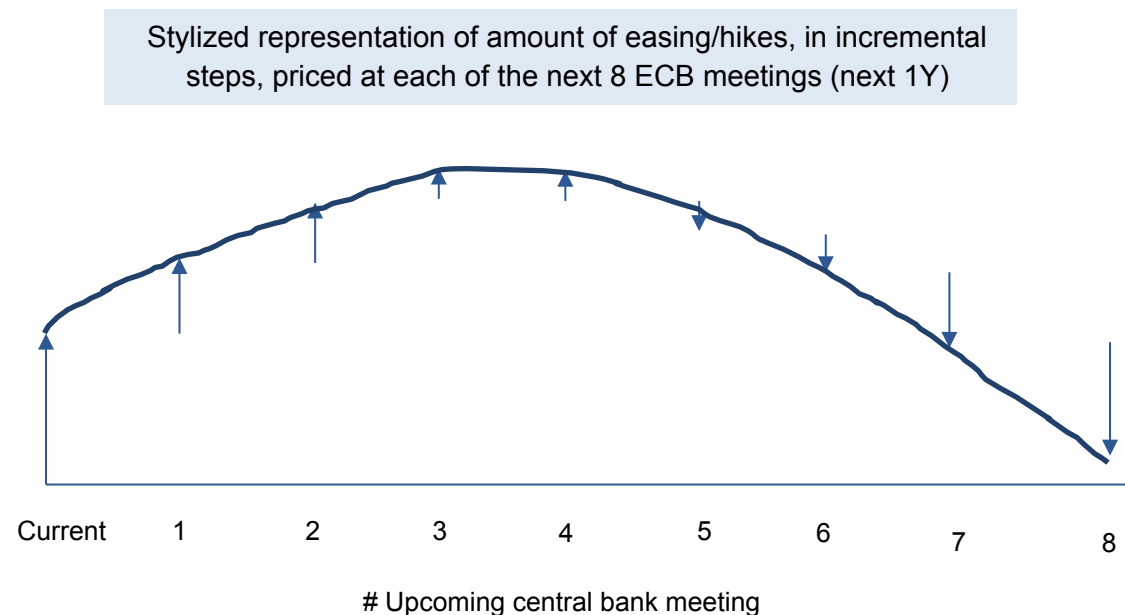
How to find fair value of the swap curve?

Step 1: Start with front-end

- Let's begin with estimating fair value at the very front-end of ECB €STR swaps
 - Primary driver: **ECB rate path expectations**



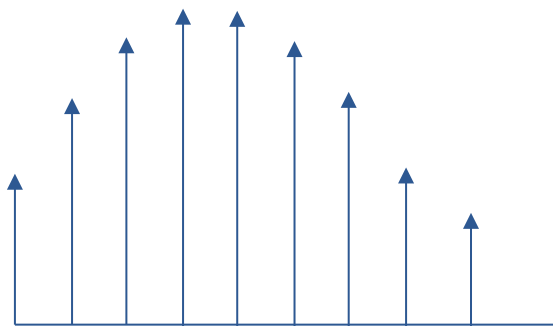
- Fair value for the next meeting: Weighted probability of outcomes: 2.15%.
- This will change with scenarios and probability associated with these scenarios.
- Current market pricing: 2.12%.
- **Trade?**



The amount of hikes/easing priced at each meeting is essentially a probability weighted sum of various outcomes.

Step 2: Modeling 1Y gaps; the forward curve should be continuous

Stylized representation of policy rate at each of the next 8 ECB meetings (1Y yield)



Interest Rate Parity:

1Y €STR swap yield can be approximated as the average ECB €STR swap yield over the next 8 meetings

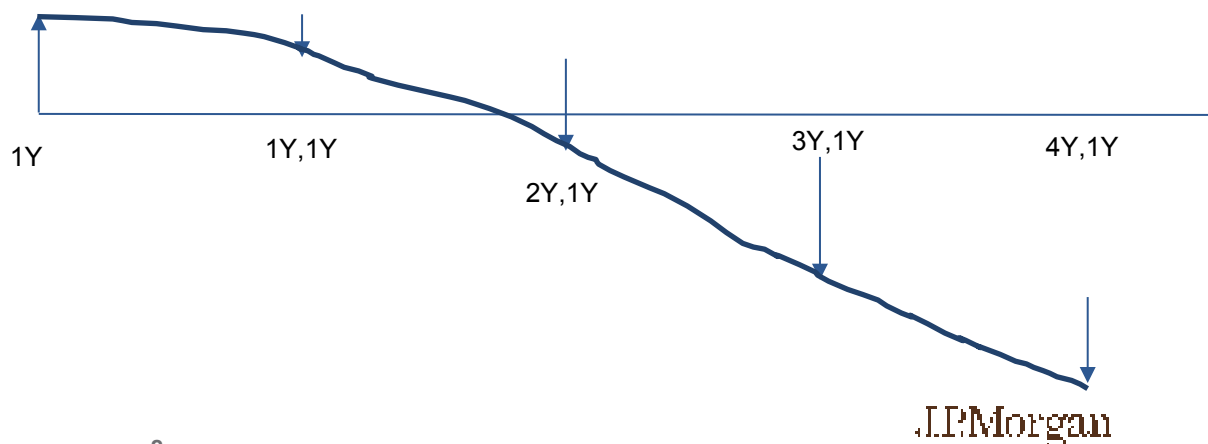
The amount of hikes/easing priced at each year gap is driven by ECB expectations over that period

How much visibility do we have on ECB policy expectations >2-3Y?

Interest Rate Parity:

- 2Y yield = average of 1Y, and (1Y,1Y)
 - (1Y,1Y) gives a leveraged exposure to 2Y swap. For each 10bp move in (1Y,1Y), 2Y swap yield will move only 5bp, assuming unchanged 1Y yield
- 5Y yield = Approximately the average of 1Y, (1Y,1Y), (2Y,1Y), (3Y,1Y), (4Y,1Y)
 - Each subcomponent has only 20% exposure to the overall 5Y rate; in other words, each forward rate offer a leveraged exposure to spot yield
 - For instance, a 5bp move in (4Y,1Y) yield would move spot 5Y rate by approximately 1bp, assuming other forwards are unchanged. In other words, if you knew all the other components except (4Y,1Y), then this component needs to be 5bp wide to for 5Y spot rate to be trading 1bp wide

The €STR curve can be built as a collection of various 1Y rates, making reasonable assumption about ECB outcomes over this period



Step 3.A: Fill in the missing parts for the long end of the curve; Term premium

- Expectations of ECB policy rates over the coming years should be the main driver of €STR fixings and consequently €STR swaps (and Euribor swaps)
- However, other factors also become relevant as we move further out
 - **Term premium**
 - Money lent out for longer should receive a premium. What are the implications of this?
 - Longer maturity warrant higher yield (linear or non-linear?) → Upward sloping curve
 - 3M Euribor fixings (indication of 3M transaction rate) should be higher than overnight fixings (€STR) and similarly 6M Euribor fixings should be higher than 3M fixings
 - OIS curve < 3s curve < 6s curve

Question: Is €STR yield < 3s yield < 6s yield?
In other words, is €STR/3s or 3s/6s basis always positive?

Step 3.B: Convexity – Pulling the long-end lower

■ Convexity

- Term structure creates a “fair” swaps rate levels (upward sloping?)
 - However, convexity adjustments drag these fair values *lower*
- Profit function of a swap is **convex** and not **linear** in yields
- P&L: Change in yield * Duration of the swap
 - If interest rates go down, then swap’s P&L is more than proportional to change in yield
 - If interest rates go up, then swap’s P&L is less than proportional to change in yield
 - This implies, lower premium needed for higher maturity

Convexity of bonds/swaps increases with maturity; real money investors prefer long-end of the curve for convexity considerations as well

Question: What is the duration of a swap?
Duration of a swap increases non-linearly with maturity – Remember bond class?

Convexity – digging deeper

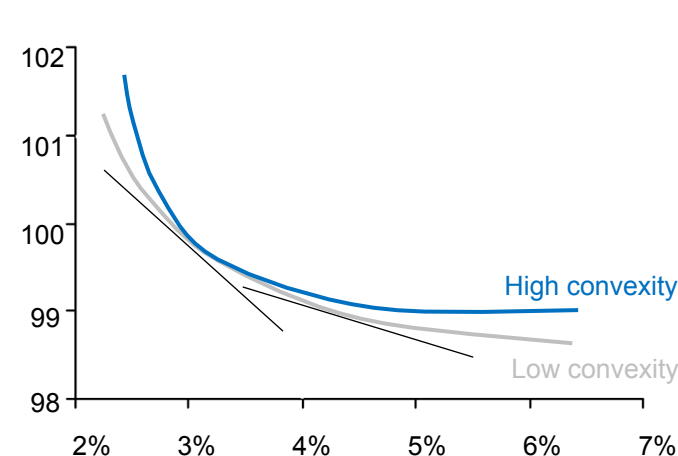
■ Linear vs. non-linear payouts (linearity with changes in interest rates)

■ Structures with **linear payouts**: Constant Maturity Swaps, Futures

- Payout == Notional * Change in interest rate
- These have constant Duration/DV01s

■ Structures with **non-linear payouts**: Bonds, swaps, options

- Payout = Notional * *Duration* * Change in interest rate
- Non-linearity arises due to price/yield convex relationship; these exist because rate of increase in bond duration decreases as yields increase → Duration/DV01 is not constant
 - For large movement in yields, bond prices increase more in rallies versus decline in sell-offs
 - This gives a cushion for yields to move higher to equate P&L for an equivalent amount rally in yields; conversely, long-end yields are “adjusted” lower due to convexity



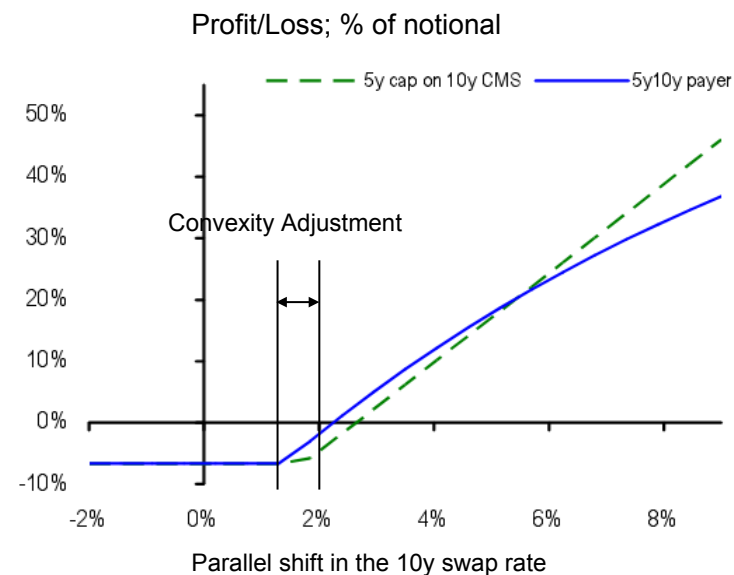
$$Cx = \frac{\sum_{t=1}^T \frac{t(t+1)CF_t}{(1 + \frac{y}{k})^{kt}}}{k^2 (1 + \frac{y}{k})^2 P}$$

Convexity adjustment is designed to produce an equal expected payoff for instruments with and without convexity

The convexity adjustment

- The convexity adjustment depends on several factors, with the key ones being:
 - Volatility, since the value of the convexity mismatch is greater when volatility is large
 - Convexity of the plain vanilla swap whose tenor matches the index – e.g., the 5-year convexity adjustment on 30-year CMS is larger than the 5-year adjustment on 3-year CMS
 - The volatility skew (e.g., the implied volatility difference between ATM and OTM payers)

Convexity adjustment is designed to produce an equal expected payoff for instruments with and without convexity



The scale of convexity adjustment

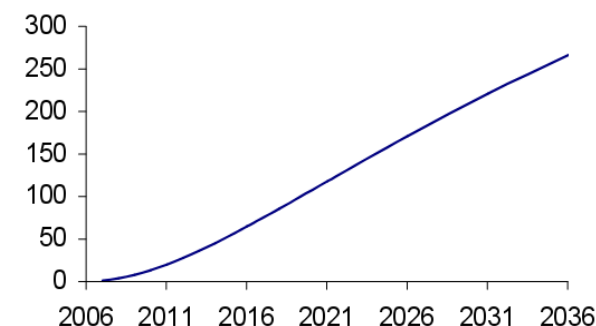
Current estimate of un-adjusted swap rate (%), CMS adjustment (bp), and adjusted CMS rate (%); as of 23rd June 2023

	Swaprate (6s) (%)	CMS adjustment (bp)	CMS rate (%)
1Yx1Y	3.51	1.4	3.52
2Yx2Y	2.82	4.3	2.86
5Yx5Y	2.71	17.8	2.89
5Yx10Y	2.77	31.2	3.08
10Yx10Y	2.59	52.4	3.11
10Yx20Y	2.22	79.8	3.02
10Yx30Y	2.02	100.6	3.03
15Yx15Y	1.96	86.0	2.82
20Yx20Y	1.64	123.5	2.87

Evolution of convexity adjustment for 10Yx10Y swap; since 1 Jan 2016; bp



Convexity vs. maturity



Convexity and hence the adjustment increases with maturity; structural demand for convexity at the ultra-long end of the curve pushes this sector further down resulting in further inversion of the curve in 30Y+

Step 3.C: Other factors

■ Flows

- Hedging flows (mortgage paying, gamma hedging, issuance related receiving, etc)
- Structural flows (Receiving needs by Pension funds/Insurance companies, etc)
- Speculative flows (RV driven?)

■ Other factors

- Policy considerations such as QE/QT, issuance dynamics, etc.
- Carry on the curve

Putting it all together – constructing a swap curve

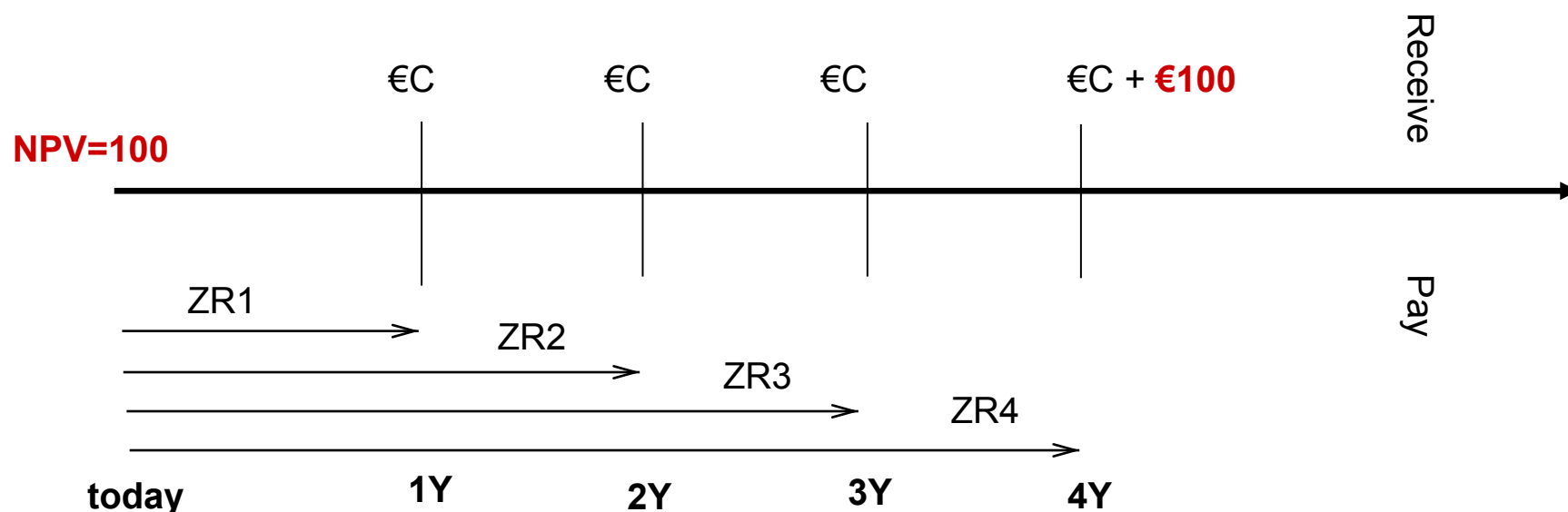
- **Expectations of ECB policy rates** over the coming years should be the **main driver of €STR fixings** and consequently €STR swaps (and Euribor swaps)
- Don't forget
 - Term premium
 - Convexity
 - Flows
 - Other esoteric factors
- Swaps using various underlying indices
 - €STR, 3s, 6s

In practice, swaps is a very liquid and deep market.
Quotes for “on-market” swaps are readily available from traders
Question: How about “off-market” swaps? (Buzzword: Bootstrapping)

Euro swaps are now cleared
through Eurex or LCH →
Eurex/LCH basis

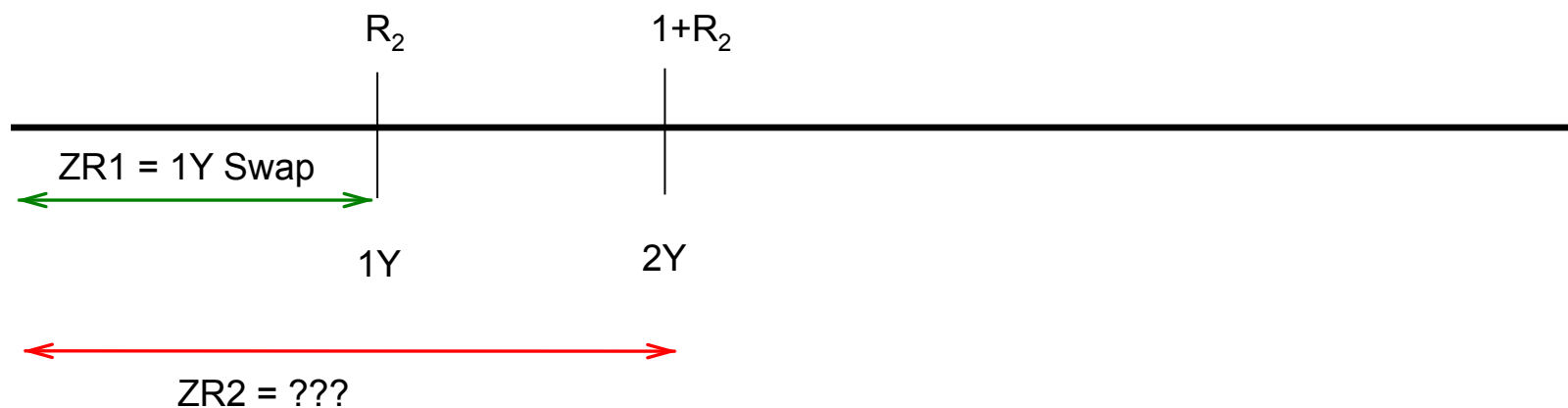
A Bond's PV is simply the sum of its discounted cash flows

- If par coupon rates are visible to us in the market...
- ...we can compute the discount rates $ZR1, ZR2, \dots$, and thereby PV any swap



Recursive calculation of zero rates (from the 1Y swap rate, 2Y swap rate, ...) is called "Bootstrapping"

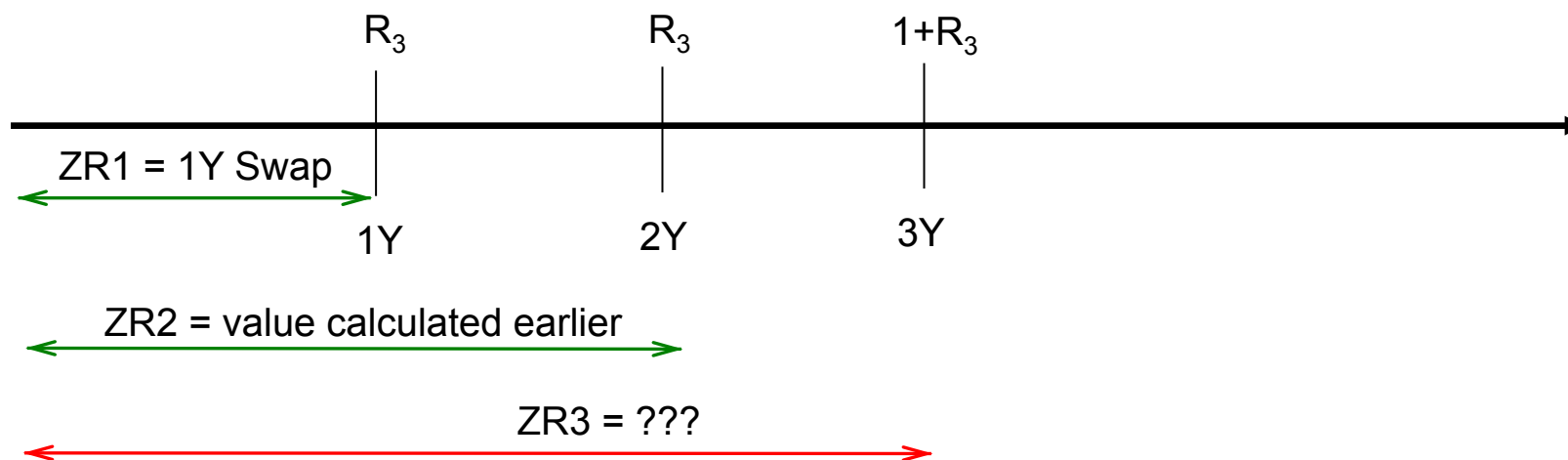
Bootstrapping...



- We know 1Y, 2Y, 3Y... swap rates (or par rates), $R_1, R_2, R_3...$
- Need to compute zero rates, $ZR_1, ZR_2, ZR_3...$
- $ZR_1 = 1Y \text{ swap rate} = R_1$
- 2Y bond with coupon R_2 is worth par today; therefore
- Solve for ZR_2

$$\frac{R_2}{1+(ZR_1)} + \frac{1+(R_2)}{(1+(ZR_2))^2} = 1$$

...demystified

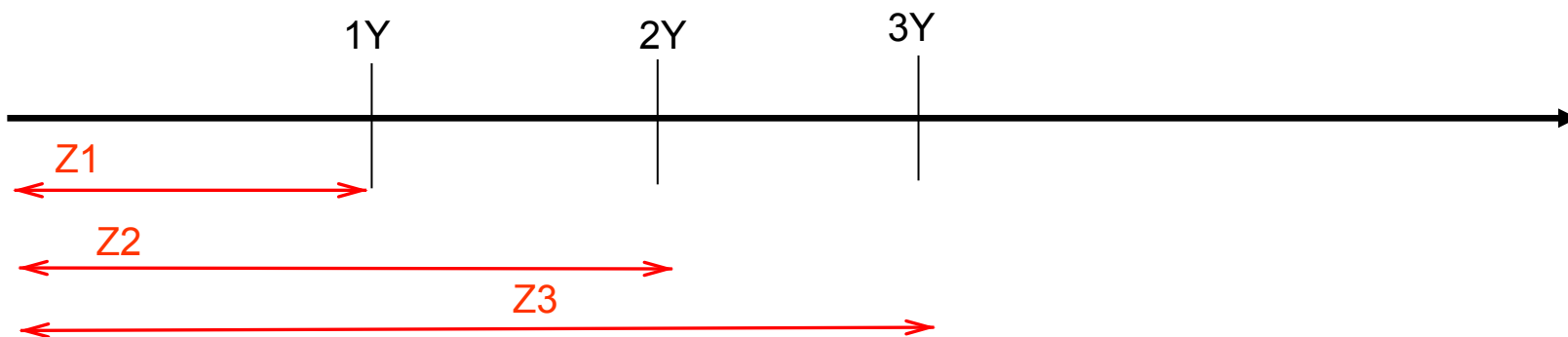


- We know $ZR1$, $ZR2$
- Let $R3$ be the 3Y swap rate
- 3Y bond with coupon $R3$ is worth par today; therefore

$$\frac{R_3}{1+(ZR1)} + \frac{R_3}{(1+(ZR2))^2} + \frac{1+R_3}{(1+(ZR3))^3} = 1$$

- Solve for $ZR3$...
- ...etc.

Bootstrapping is easier if we first find discount factors rather than discount rates ...



- Suppose R_1 , R_2 and R_3 denote 1Y, 2Y and 3Y swap yields...
- ...then:

$$(1+R_1) \times Z1 = 1$$

$$R_2 \times Z1 + (1+R_2) \times Z2 = 1$$

$$R_3 \times Z1 + R_3 \times Z2 + (1+R_3) \times Z3 = 1$$

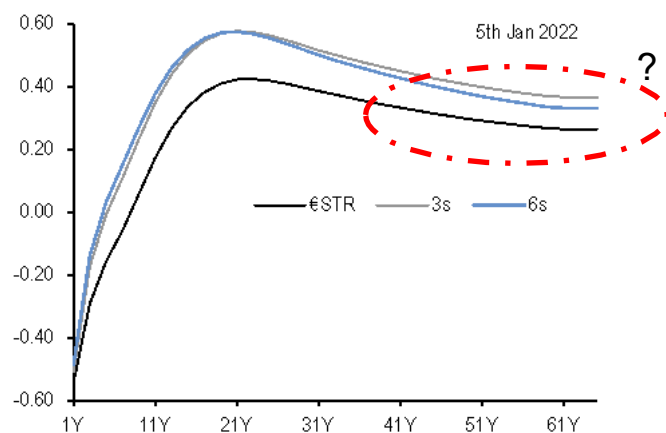
- We can solve for 3 unknowns with 3 equations
- Once discount factors are estimated, any off-market swap can be valued

Dual discounting adds a further layer of complexity

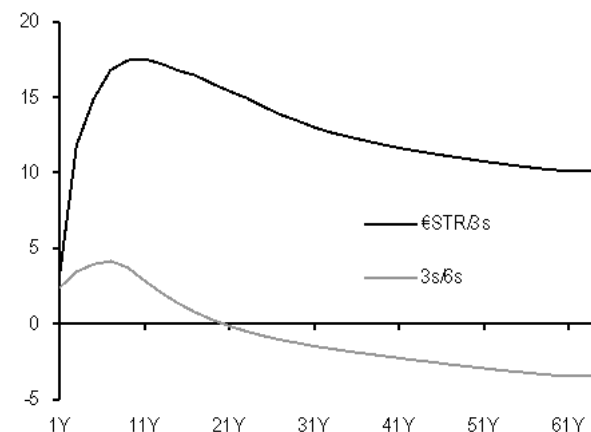
- Further complexity in the pricing of swaps resides in projection curve being not equal to discounting curve
 - Projection cash-flows are calculated on the –IBOR curve; these are the projected cash-flows over the life of the swaps
 - These cash-flows are discounted on the OIS curve, as collateralised swap can be hedged via OIS swaps
 - The PV of a ‘market’ swap still remains zero, as the floating and the fixed leg of the swap have the same values, however the value of each of the fixed and of the floating legs is no longer 100

Basis curves

Shape of EUR swap curve; 6M Euribor, 3M Euribor, and €STR; 5th Jan 2022; %



Euribor/€STR and 3s/6s basis; 5th Jan 2022; bp



■ Main drivers:

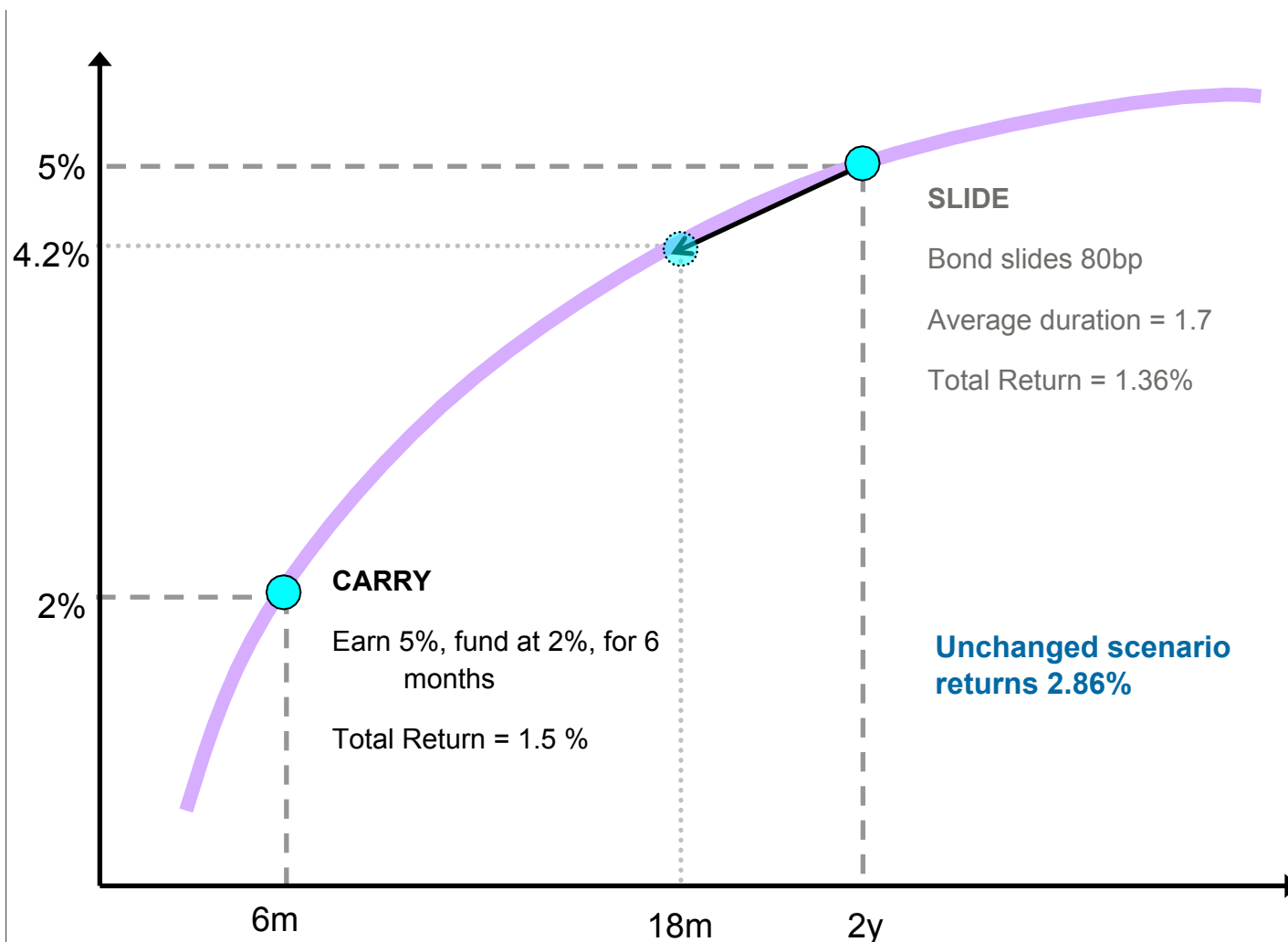
■ Structural flows

- PFs receive in 6s because the discount using the 6s curve
- 6s trade below 3s at the ultra-long end; 20Yx10Y 3s/6s basis is currently -13bp

■ Libor transition

- Fixes these spreads depending on overnight spot rates

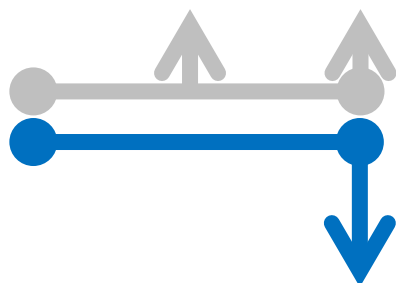
Carry and Slide returns in a positively sloped yield curve



- **Carry** is real earning (return vs funding)
 - A big but should not be the only decision making criteria
 - Should be looked at versus the risk of asset price movement
- **Slide** is conceptual (assuming curves remains unchanged)

Carry on swaps may be computed from forward par rates

Carry: Coupon income less financing cost from holding the trade



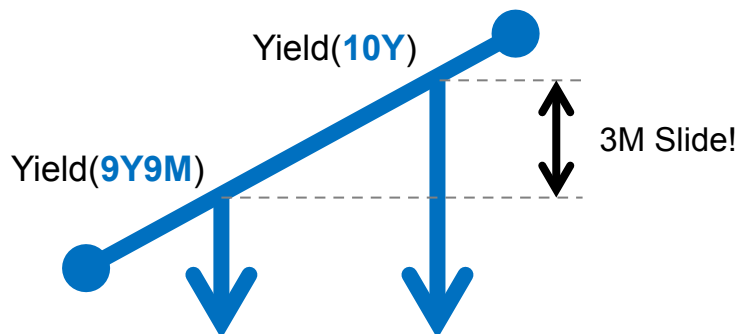
$$\text{Yield}(\mathbf{10Y}) + \text{Carry}(\mathbf{3M}) = \text{Yield}(\mathbf{3M \times 9Y9M})$$

(from no arbitrage)

$$\text{Carry}(\mathbf{3M}) = \text{Yield}(\mathbf{3M \times 9Y9M}) - \text{Yield}(\mathbf{10Y})$$

(subtraction)

Slide: Expected gain/loss from rolling down the yield curve



$$\text{Slide}(\mathbf{3M}) = \text{Yield}(\mathbf{10Y}) - \text{Yield}(\mathbf{9Y9M})$$

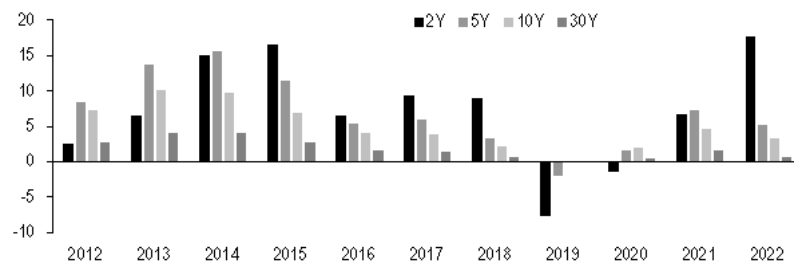
Carry on swaps may be computed from forward par rates

- Carry + Slide is loosely referred to as just carry, and frequently thought of as the cost of holding a trade

$$\frac{\text{Yield}(1Y \times 9Y) - \text{Yield}(10Y)}{\text{carry}} + \frac{\text{Yield}(10Y) - \text{Yield}(9Y)}{\text{slide}} = \frac{\text{Yield}(1Y \times 9Y) - \text{Yield}(9Y)}{\text{carry + slide}}$$

- An investor receiving fixed in a 10-year swap today (or equivalently, in a 1Yx9Y forward starting swap) expects to make the carry over a year if the spot yield curve remains unchanged
- Thus, being long 10s is said to carry positively, while shorts are negative carry trades
- It is common to think of carry + slide as breakeven moves in spot yields; to think of it as projected P/L is a stretch

3M carry and slide in various swaps as of Jan each year (bps)



Source: JPMorgan

Myriad users of Interest Rate Swaps: Swaps are among the most popular derivatives

Swaps are mainly used for:

■ Portfolio Management

- Adjust interest rate exposure and offset risks posed by interest rate volatility; express views on credit spreads and/or less liquid fixed income instruments

■ Corporate finance/Rate-locks on bond issuance

- Shift risk exposure from fixed rate liability to floating or vice versa
- Rate-locking of issued fixed rate bonds by entering into swaps. That gives them time to find investors for bonds. Once they find investors, they sell the bond and unwind swap hedges

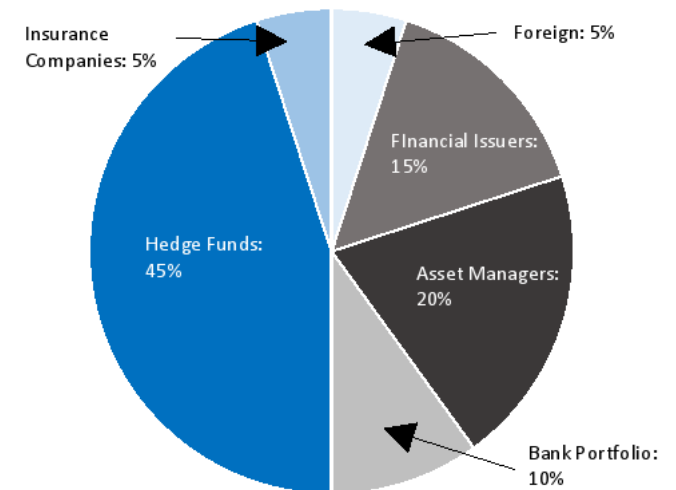
■ Risk management

- Banks and other financial institutions manage residual interest rate risks via swaps

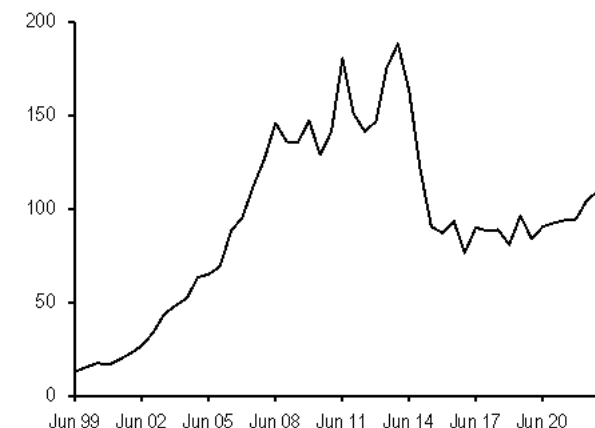
■ Speculation

- Speculate movements in interest rates curve/shape/curvature. NPV=0 makes it attractive.

Swap market participants (estimated)



Total notional outstanding for EUR swaps; since Jan 1999; \$tn



Swap flows – who are payers/receivers of Euro IRS: Sub 10Y

Payers of fixed in belly (5Y-10Y)

■ Banks

- Typically long assets via bonds; hedge interest rate risk via paying in swaps
- Prefer paying in 3s and through Eurex

■ Corporates

- Prefer paying in 6s when rates are low to pre-hedge issuance

■ Treasury flows

- Rate locking flows
- Moving towards €STR from historically 3M Euribor

■ Issuance pre-hedge/rate lock

- Some SSAs pay vs. 3M ahead of issuance to lock levels and hedge against market moves

■ Yankee Issuance

- EUR names issuing in USD and swapping back to EUR; need to eventually pay fixed in swaps
- Cross currency trading is done on a RFR basis but these flows are typically swapped back to 6s

Receivers of fixed in belly (5Y-10Y)

■ Issuance

- Issuers from corporates (mainly financials) swap their fixed rate liabilities into floating by receiving fixed in swaps; mainly 5-7Y
- Starting to see more swapping vs €STR as opposed to 3s (or some vs 6s)

■ SSAs

- Generally prefer receiving vs 3M Euribor

■ ALM flows

- Shorter-dated deposit hedging from the passive side; reduced deposits if interest rates are low; receive in swaps to protect against this scenario

Swap flows – Long-end (10Y+)

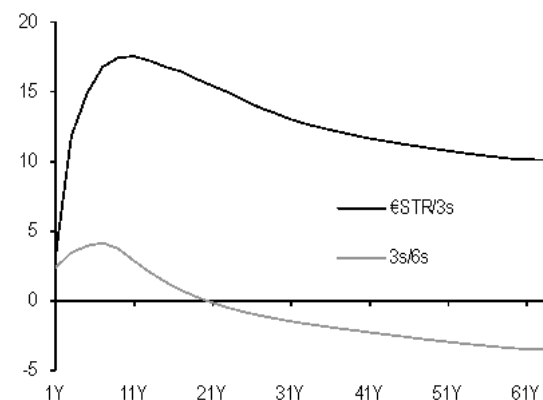
Payers of fixed (10Y+)

- **Buyers of syndications/SSAs**
 - Typically done via ASWs on €STR
 - E.g., EU/OAT syndication
- **RM buyers of long end EUR cash**
 - Typically done via ASWs on €STR
- **Ultra long-end rate locking flows**
 - Continental European RM players
- **Insurance hedging**
 - Less frequent; done on 6s

Receivers of fixed in belly (10Y+)

- **Pension funds**
 - Hedge mismatch durations of assets vs liabilities
 - Usually via 6s and Eurex/LCH mix
- **Insurers**
 - Liability hedging
 - Almost always on 6s
- **Issuance receiving**
 - Typically on 3s; mainly from Finnish corporates

Notable impact on long-end tenor basis from flows



Using swaps: Pension funds – market data

Assets		Liabilities	
Real Estate	177	Own funds (Total)	233
Equities	566	Specified Reserve	109
Alternative Investments	136	Other	123
Fixed Income Securities	893	Technical provisions (Total)	1592
Hedge funds	26	Fund's risk	1578
Commodities	6	Guarantee contract	6
		Member's risk	8
Other	33	Others	12
Total	1836	Total	1836

* Data as of YE 2021

■ Duration:

- Duration of PF liabilities are generally much larger relative to their duration of assets
 - Equity holdings have zero duration
 - Other asset durations are likely less than liability duration (>20Y)

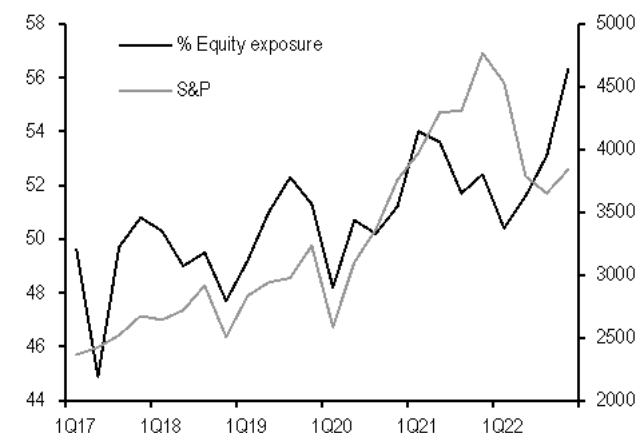
■ Coverage Ratio:

- Defined as (NPV of Assets/NPV of Liabilities); ideally should be >100%

■ Hedge Ratio:

- Ratio of Interest Rate exposure of Assets to Liabilities

Evolution of APF's % exposure to equity investments (lhs) % vs S&P (rhs)

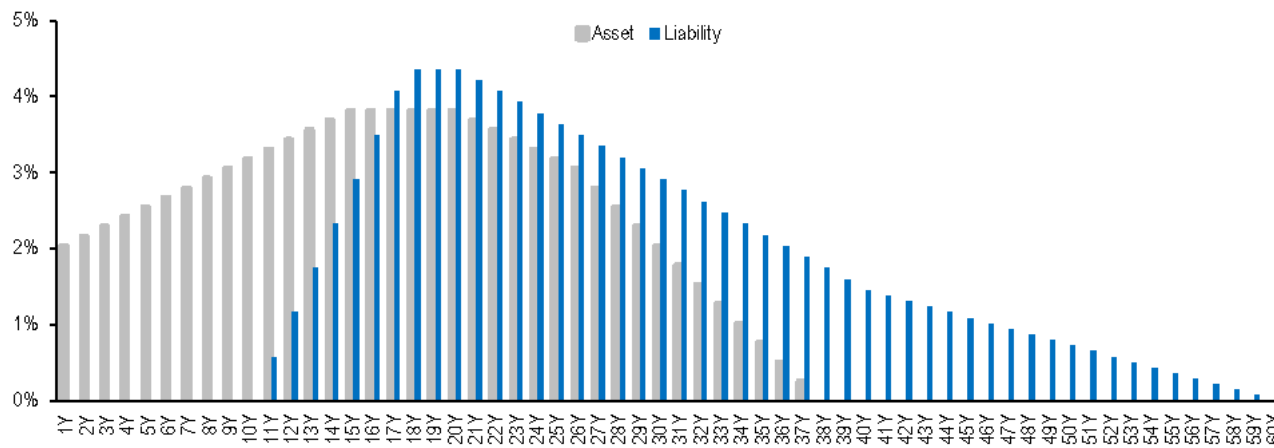


Evolution of Dutch PF industry coverage ratio; %



Pension funds: Low rates pose a risk to coverage ratios; receive in swaps

Stylized representation of distribution of duration of a typical PF's Assets and Liabilities; %

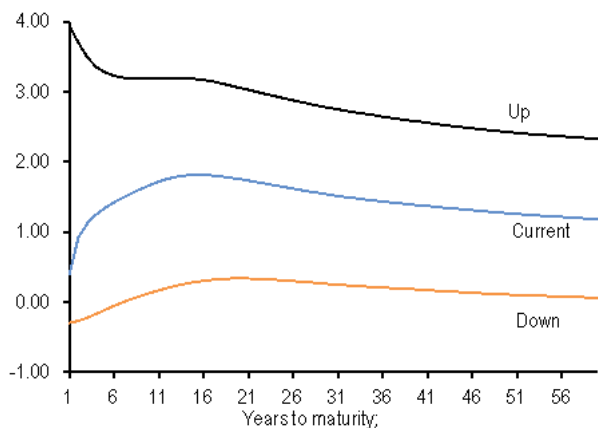


Remember:

$$\text{Coverage Ratio} = \frac{\text{NPV of Assets}}{\text{NPV of Liabilities}}$$

- Duration of Liabilities > Duration of Assets
- Discounting curves play a crucial role in estimating the NPV
 - Think Dutch PF reforms impact on the discounting curves

What happens to Assets and Liabilities NPV in large interest rate moves?

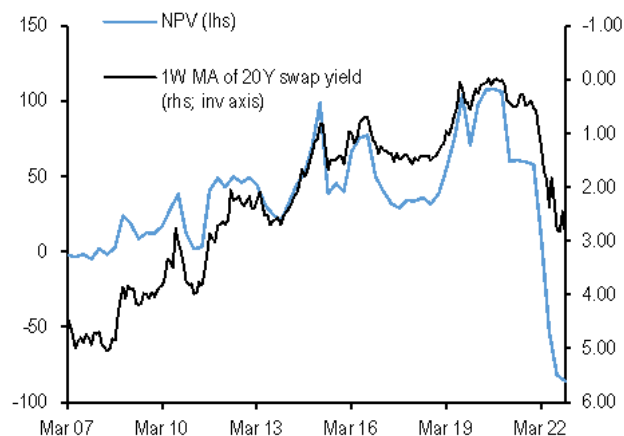


Yield move	Current	Up	Down
NPV Assets		↓	↑
NPV Liabilities		↓	↑
Coverage Ratio	100%	>100%	<100%

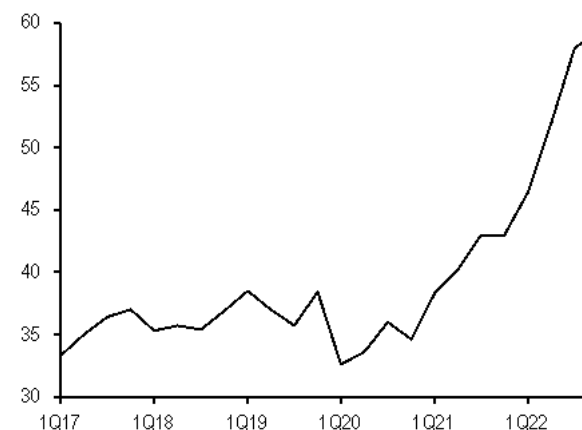
Lower rates is a risk for PFs → **Structural need to receive in long-end rates**

Dutch pension funds regulatory changes and its impact

Evolution of NPV of Derivatives position (lhs) versus 20Y swap yield (rhs; inv axis)



Actual interest rate risk exposure hedged for APF; %



- Interest rates risk are not fully hedged by a typical PF
- **Dutch PF** industry is a big player at the long-end of the EUR swap curve; natural bias is towards receiving in 30Y+ sector. A reduction of these flows should support higher long-end rates, all else equal
- Recent Dutch PF reforms looks to push the last liquid point (LLP) from a market perspective from 30Y to 50Y → Some curve steepening bias
- Switch from Defined Benefit (DB) to Defined Contribution (DC) schemes is likely to require less hedges → Some 10s/30s curve steepening bias, if hedges in 30Y sector are unwound

Using swaps: Insurance companies

- **Money in Sooner / Money out later**
- **Liabilities** (“Money Out”):
 - Lump sums or annuities
 - These future long-dated cash flow requirements (30Y+) deliver duration exposure
- **Assets** (“Money In”): Invested to meet these forward liabilities
 - Generally invested in fixed income assets to match cash flows and/or offset duration from liabilities
 - Typical assets: Government bonds + Credit
 - Constraints with using fixed income bonds:
 - Long-dated core EGBs do not give the desired return to shareholders
 - Discount curve used to value liabilities is primarily IRS based and surplus capital requirement is based of this. Thus, OW EGBs creates volatility in this surplus capital
 - Credit market is relatively illiquid beyond the 10Y point
 - Long-dated credit suffers from high credit charges

- Remember:

$$\text{Solvency Ratio} = \frac{\text{NPV of Assets} - \text{NPV of Liabilities}}{\text{Solvency Capital Requirement}}$$

- SCR is essentially the “own equity” of the company (Total assets – total liabilities). This is also interpreted as the amount of money that the company needs to reserve to compensate for market risk and Life Underwriting risk
- Unlike PFs, Inco’s are managed via their “Solvency Ratio”. They are required to hold capital called SCR (Solvency Capital Requirement)
- Solvency Ratio must be > 100%

Insurance companies – main players in EUR IRS

- Primary EA countries who participate in swap hedging activity:
 - **Netherlands:**
 - Very long dated liabilities
 - Insufficient yield from Dutch / German govies, and insufficient longer-dated credit
 - **France:**
 - Shorter-dated liabilities (7-10Y duration) but still find swaps a convenient way to manage duration
 - **Italy:**
 - Shorter-dated liabilities (7-10Y duration); Overweight BTPs and need more swap hedging
 - **Denmark:**
 - Long-dated liabilities and no domestic assets; use EUR swaps as a way to source a significant amount of duration in EUR

Using swaps: Banks

■ Hedge against losses (e.g., loan portfolio)

- Values of fixed-rate loans and Investments in fixed rate securities declines when rate rises; pay swaps to hedge this risk
- Large scale paying in intermediate sector swaps in 2022 from German and French banks to hedge their loan book
- At an aggregate, around 80-85% of European mortgages are fixed rate

Hedge Fixed assets by paying

■ Manage interest rate risk

- NPV of liabilities (such as long-term fixed rate bond issuance) increase as interest rate decline; hedge against this via receiving in swaps
- Turn floating rate assets into fixed by receiving in swaps

Hedge long-term liabilities, floating rate assets by receiving

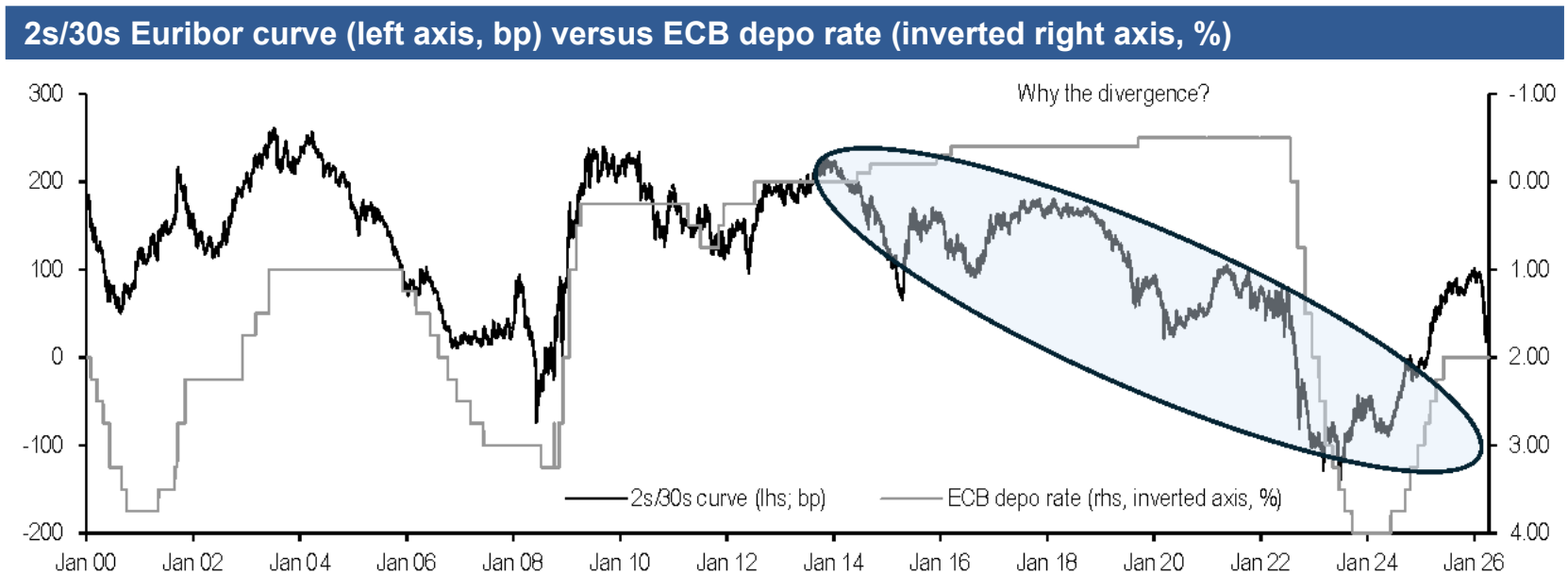
■ Speculate?

- Receiving/Paying based on expected interest rate cycle (macro/structural) or event driven moves (tactical)

Snap shot of J.P.Morgan balance sheet; \$bn

Assets		Liabilities	
Loans (consumer)	500	Deposits	2313
Loans (commercial)	640	Demand	670
Interbank Assets (Fed funds, repo, other)	1040	Interest Bearing	1641
Investments (ST and LT)	1015	Short Term borrowing / Repo	375
Trading securities / Available for Sales (AFS) / Hold-to-maturity (HTM)		LT Debt	315
Other Assets (fixed, Intangible, Derivative)	300	Other Liabilities	281
Cash	28	Capital (Tier 1)	292
Intangible assets	53	Retained Earnings / Share Capital / Pref equity minus 'other' and 'Tsy stock'	
Minus reserves for loan losses.		Minus reserves for loan losses.	
Total	3576	Total	3576

Trading the yield curve: What drives its shape?

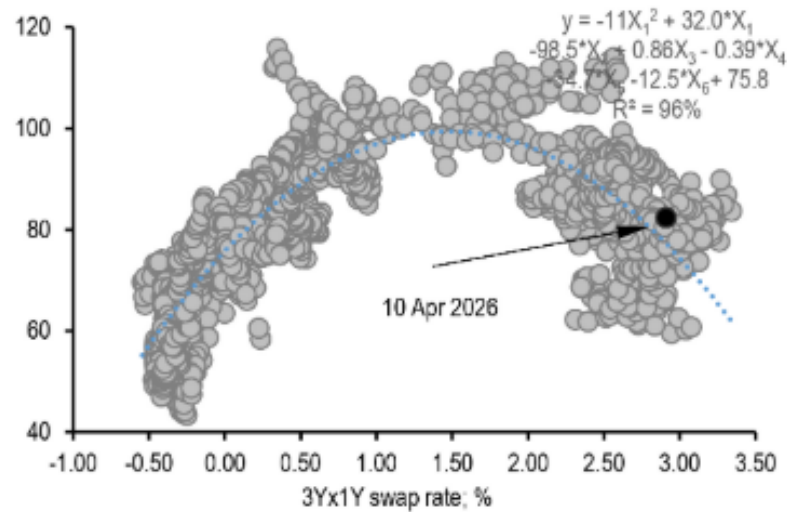


Source: JPMorgan

Medium-term steepening bias on the 10s/30s curve

10s/30s curve has flattened substantially over the past few weeks and now appears broadly fair value versus its fundamental drivers...

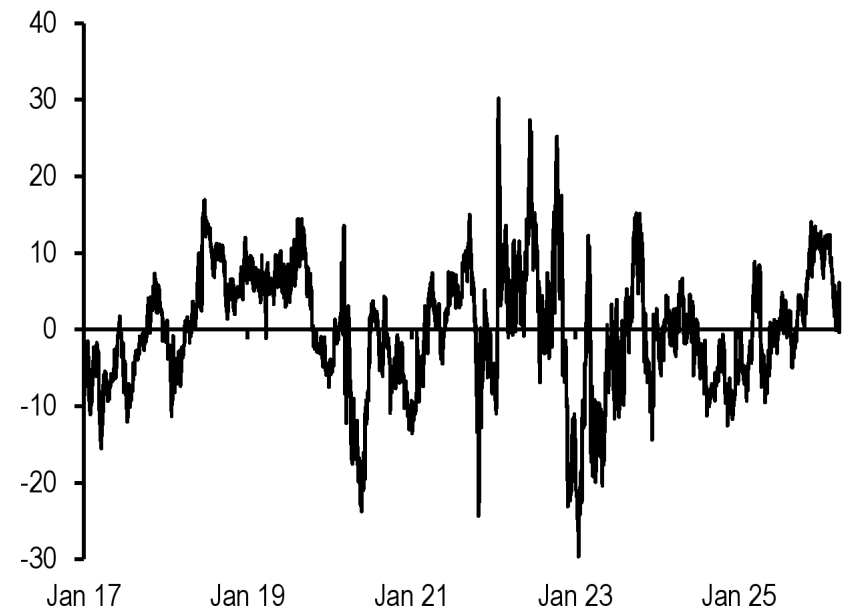
10s/30s swap curve regressed against 1) 3Yx1Y swap rate (X_1 , quadratic fit), 2) ECB balance sheet as a % of GDP (X_2), 3) 10s/30s USD swap curve (X_3), 4) VIX (X_4), 5) ECB Jan22 meeting dummy and Fed's Dec23 meeting dummy (X_5 & X_6); since 1 Jan 2017; bp



Source: J.P. Morgan.

...potentially indicating that steepeners that were implemented in anticipation of the Dutch PF transition flows are likely clean now

Residual front the adjacent regression; since 1 Jan 2017; bp



Source: J.P. Morgan.

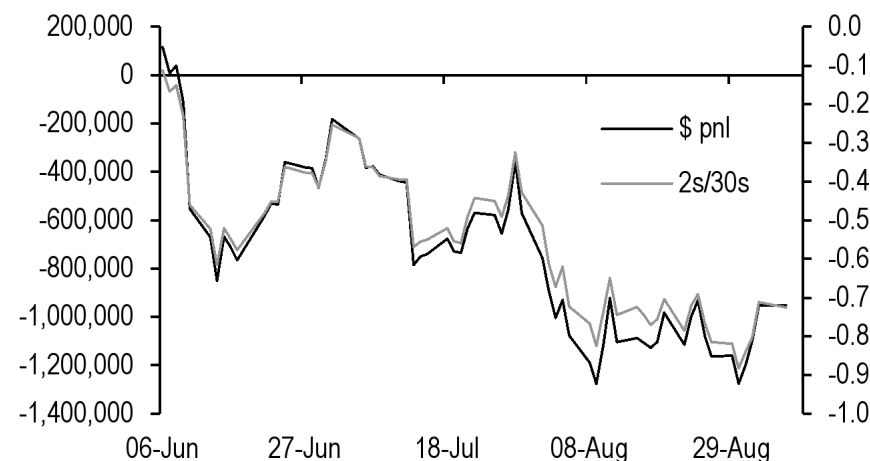
Trading the curve: an example

Swap curve steepener

Market Condition:

- \$100mn notional 2Y receive-fixed swap
 - Swap yield: 2.796%
 - Swap start: 6/7/2022
 - Swap end: 6/7/2024
 - PVBP: \$198/bp per mn notional
- \$X mn notional 30Y pay-fixed swap
 - Swap yield: 2.63%
 - Swap start: 6/7/2022
 - Swap end: 6/7/2052
 - PVBP: \$1966.9/bp per mn notional

Curve steepener pnl (lhs, \$) versus 2s/30s swap curve (rhs, bp)



Source: JPMorgan

We want X such that the whole package is **neutral to parallel shifts in the curve**
that is $\$100\text{mn} * \$198/\text{bp per mn notional} = X * \$1966.9/\text{bp per mn notional}$

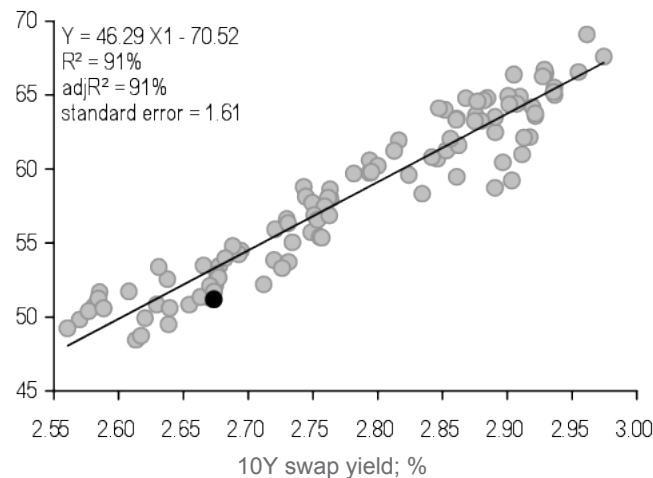
$$X = \frac{198}{1966.9} * 100 = 9.7$$

So to construct a swap curve steepener, we want to receive \$100mn 2Y swaps and pay \$9.7mn 30Y swaps

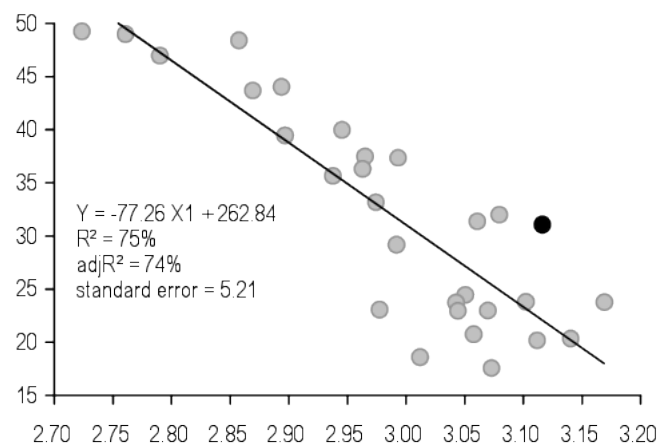
JPMorgan

Curve trades as alternative ways of trading the market

2s/10s EUR swap curve versus 10Y swap yield; 1st Oct 2025- 1st March 2026; bp



2s/10s EUR swap curve versus 2Y swap yield; since 1st March 2026; bp



- Investors who expect the market to rally often position in curve steepeners
- Investors who expect the market to sell-off often position in curve flatteners
- The right curve points can provide an extra edge on a macro trade
- Find points which look too flat or too steep given the level of yields
- Or buy the points which carry positively, and be short at points with low or negative carry

What should you do when you don't have a view on the direction of yield move, but strong conviction on the yield curve directionality?

Disclosures

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Company-Specific Disclosures: Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan’s implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

Disclosures

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC (“JPMS”) acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries (“J.P. Morgan Private Bank”), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina (“BCRA”- Central Bank of Argentina) and Comisión Nacional de Valores (“CNV”- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited (“JPMSAL”) (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Disclosures

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P. Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC).

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai - 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH0000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number - INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpmypl.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Prasanna Bandal; prasanna.bandal@jpmchase.com; +912261575159. Grievance Officer: Ramprasad K, jpmypl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](http://www.jpmypl.com/#research). The annual Compliance audit report is available at <http://www.jpmypl.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V. and J.P. Morgan Grupo Financiero are members of the Mexican Stock Exchange and are authorized to act as a broker dealer by the National Banking and Securities Exchange Commission.

Disclosures

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission, and its registered address is 3rd Floor, 20 North Sathorn Road, Silom, Bangrak, Bangkok 10500.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPMML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for

Disclosures

any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised April 04, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

J.P.Morgan